

An Introduction to Mean-Variance Portfolio Optimization

Theory, Modeling, and Convex Quadratic Programming

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Abstract

A portfolio optimization problem consists of allocating capital among a set of assets in order to minimize risk while achieving a desired expected return, subject to budget constraints. The mean-variance framework introduced by Markowitz formalizes diversification by modeling returns through their expected value and covariance structure.

Random returns are modeled as a vector $r \in \mathbb{R}^n$, with mean $\mu = \mathbb{E}[r]$ and covariance matrix Σ . Let $w \in \mathbb{R}^n$ denote the portfolio weights. The expected return of the portfolio is $\mu^T w$, and the risk is measured by the variance $w^T \Sigma w$.

The problem can be formulated as a convex quadratic program:

$$\begin{aligned} \min_{w \in \mathbb{R}^n} \quad & w^T \Sigma w \\ \text{s.t.} \quad & \mu^T w \geq r_{\text{tar}}, \\ & \mathbf{1}^T w = 1, \\ & w \geq 0. \end{aligned} \tag{1}$$

We also discuss extensions such as cardinality constraints and additional restrictions on asset allocation. A practical example using S&P 500 data is presented, including estimation of model parameters and numerical optimization.

1 Introduction

Suppose you are an investor, and you are interested in having a non-decreasing fund, even when the market is losing value. This is generally not possible when investing in a single asset. In the financial market, it is rare for all securities to gain or lose value at the same time, therefore an investor should use a diversification strategy. Diversification reduces risk by combining assets whose returns do not move perfectly together. When assets are not perfectly correlated, losses in some positions may be counteracted by the gains in others, resulting in overall better portfolio returns. The goal is then to decide the weights of the investment.

One way to model the problem is Markowitz's Mean-Variance Portfolio Theory, a theory based on the portfolio optimization problem developed in Harry Markowitz's 1952 paper *Portfolio Selection*. The problem can be stated as follows:

Consider n assets or stocks held over a period of time. Let μ be the expected returns of the assets, and Σ the covariance matrix, which gives the volatilities of the asset returns and the correlation among them. We will consider the situation where the investor does not allow short-selling.

Short-selling refers to the practice of borrowing and selling an asset with the expectation of repurchasing it later at a lower price. In this case, the investor takes a negative position in the asset, and so we will consider solely weights greater or equal to 0.

An investor desires to achieve the maximum expected return for a portfolio while keeping the risk below a certain threshold. With convex programming we can express the problem as

$$\begin{aligned} & \max(\mu^T w) \\ \text{Subject to: } & w^T \Sigma w \leq (\sigma_{tar})^2 \\ & \sum_{i=1}^N w_i = 1 \\ & 0 \leq w_i \end{aligned}$$

where $w \in \mathbb{R}^n$ are the portfolio weights, and σ_{tar} is the target portfolio standard deviation or volatility.

This problem is a convex quadratic program, as the objective function is convex and the feasible set defined by the constraints is also convex.

There are many other ways to formulate the trade-off between expected return and risk as an optimization problem. An equivalent problem would be to minimize the return variance subject to achieving a minimum acceptable mean return μ_{min}

$$\begin{aligned} & \min(w^T \Sigma w) \\ \text{Subject to: } & (\mu^T w) \geq \mu_{min} \\ & \sum_{i=1}^N w_i = 1 \\ & 0 \leq w_i \end{aligned}$$

These two formulations describe the same fundamental trade-off between risk and return. By varying the target return or the risk bound, one can obtain different optimal portfolios corresponding to different levels of risk tolerance.

An alternative and widely used formulation combines the two objectives into a single expression:

$$\begin{aligned} & \min(w^T \Sigma w) - \lambda(\mu^T w) \\ \text{Subject to: } & \sum_{i=1}^N w_i = 1 \\ & 0 \leq w_i \end{aligned}$$

where the parameter $\lambda > 0$ controls the trade-off between risk and return. Larger values of λ place more emphasis on expected return, while smaller values prioritize risk minimization.

It is clear to see that the solution of our problem fully depends on the mean and covariance matrix, and ultimately on how we determine their value in practice. In general, portfolio modeling requires two steps: data modeling and optimization. In Section 2 of this paper we discuss the nature of financial data and the estimation of the mean and covariance. Then in Section 3 we discuss the convex optimization problem (COP) and extensions of the model. Finally, in Sections 4 and 5 we demonstrate the portfolio selection with historical data from the S&P 500, and discuss the results.

2 Background

3 Financial data

3.1 Prices and Returns

In portfolio optimization, for our purposes, the primary quantity of interest is the return of an asset, as investment decisions are ultimately driven by the objective of maximizing returns. Returns r_i are derived from the price of an asset over time, *“the most obvious quality one can observe in financial markets”*, which we denote by p_t , and the sequence $\{p_t\}$ can be viewed as a stochastic process or time series.

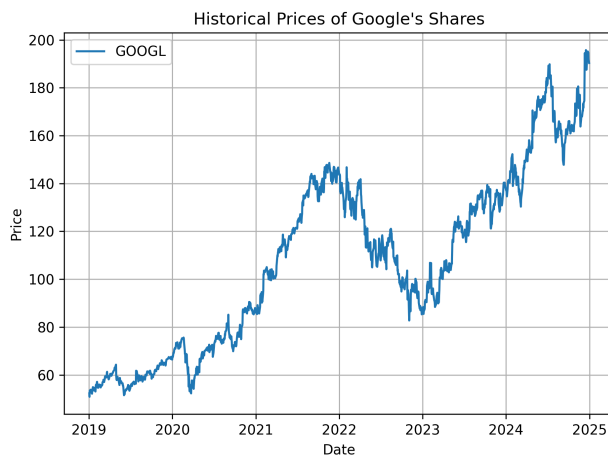


Figure 1: Historical share prices of Google stock, from 2019 to 2025

Returns measure the relative change in price(p_t) between two time periods (e.x. minutes, hours, days, weeks, quarters, or years).

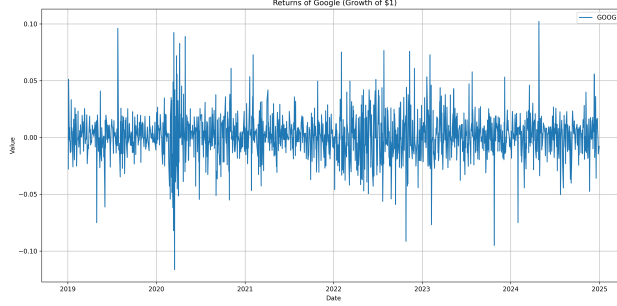


Figure 2: Google's returns

The linear return is defined as

$$r_t^{\text{lin}} = \frac{p_t - p_{t-1}}{p_{t-1}} = \frac{p_t}{p_{t-1}} - 1. \quad (2)$$

Linear returns have the convenient property that portfolio returns can be expressed as the weighted sum of the individual asset returns.

Sometimes it is more practical to work with the logarithm of prices, defined by

$$y_t = \log p_t. \quad (3)$$

In this case, the log-return is given by

$$r_t^{\text{log}} = y_t - y_{t-1}. \quad (4)$$

The relationship between linear and log-returns is

$$r_t^{\text{log}} = \log(1 + r_t^{\text{lin}}). \quad (5)$$

A key characteristic of returns is their variability, commonly measured by the standard deviation, also known as volatility. Assets with annualized volatility between 12% and 20% are typically considered to have low to moderate volatility, while values above 30% indicate high volatility.

3.2 Estimating μ and Σ

The modeling framework we adopt follows the standard i.i.d. approach commonly used in financial data analysis (see Palomar, 2025). Under the efficient-market hypothesis, asset prices are assumed to incorporate all available information, so that future price changes are driven only by new information and are therefore difficult to predict. This motivates modeling prices as a random walk and, correspondingly, treating returns as random variables with no predictable temporal structure.

For multiple assets, returns are modeled as a multivariate random variable. Under simplified assumptions, these returns are often considered independent and identically distributed (i.i.d.), meaning that returns at a given time are independent of past observations and their distribution remains constant over time.

Under this assumption the returns are modeled as

$$r_t = \mu + \epsilon_t \tag{6}$$

where $\mu \in \mathbb{R}^N$ denotes the expected return and $\epsilon_t \in \mathbb{R}^N$ is the residual with zero mean and covariance matrix $\Sigma \in \mathbb{R}^{N \times N}$.

We estimate (μ, Σ) , which are unknown in practice, using historical data of the past T observations $r_1, \dots, r_T$.

The simplest estimator is the sample mean

$$\hat{\mu} = \frac{1}{T} \sum_{t=1}^T r_t$$

and the sample covariance matrix

$$\hat{\Sigma} = \frac{1}{T-1} \sum_{t=1}^T (x_t - \hat{\mu})(x_t - \hat{\mu})^T$$

. These estimators are unbiased and consistent.

While the i.i.d. assumption is simple and convenient, it is an approximation. In practice, financial data often show patterns over time, such as periods of high and low volatility, and other approaches also suggest that markets may not always behave in a fully efficient way. Since we will not be considering these features of the data, for now, when working with real data, the results may be less reliable, since the modeling assumptions may not hold exactly in practice.

See Palomar 2025 for a more in-dept discussion of these assumptions.

4 COP & Model Extensions

We consider the mean-variance portfolio optimization problem introduced in Section 1, where the objective is to maximize expected return subject to a risk constraint and no short-selling.

$$\begin{aligned} & \max(\mu^T w) \\ \text{Subject to: } & w^T \Sigma w \leq (\sigma_{tar})^2 \\ & 0 \leq w_i \end{aligned}$$

In practice, portfolio optimization problems often include additional constraints that better reflect real-world considerations. These may be necessary to reflect transaction costs, regulatory requirements, or the need to limit exposure to specific assets. In this section We present some of the most common.

Budget Constraint

$$\sum_{i=1}^N w_i \leq 1$$

The budget constraint ensures that the total amount invested does not exceed the available capital. Allowing the sum to be less than one corresponds to keeping a portion of the capital uninvested.

Diversification Constraint

$$l \leq w \leq u$$

The diversification constrain (the upper bound) limits the maximum positions to avoid overexposure and diversify. Sometimes a lower limit l is also constraints, in the case where it is desired to hold some assets.

Cardinality Constraint

$$||w||_0 \leq K$$

The cardinality constraint limits the number of assets that can be included in the portfolio to at most K . This reflects practical situations where an investor prefers to manage a limited number of assets due to transaction costs or simplicity.

Leverage Constraint

$$\sum_{i=1}^N |w_i| \leq L$$

The leverage constraint limits the total exposure of the portfolio. When short-selling is allowed, this constraint controls how large the absolute positions can be, preventing excessive borrowing or risk-taking.

4.1 Efficient Frontier

The solutions to the mean-variance optimization problems can be represented graphically through the *efficient frontier*. Each feasible portfolio corresponds to a point in the risk–return plane, where the horizontal axis represents the portfolio risk (typically measured by the standard deviation $\sqrt{w^T \Sigma w}$) and the vertical axis represents the expected return $\mu^T w$.

The efficient frontier is the set of portfolios that achieve the highest expected return for a given level of risk, or equivalently, the lowest risk for a given level of expected return. Portfolios lying below the frontier are suboptimal, as there exist other portfolios with higher return for the same risk or lower risk for the same return.

If we vary the risk constraint σ_{tar} , or equivalently the target return μ_{min} , we can draw the efficient frontier (Fig. 4.1), which shows the trade-off between risk and return faced by the investor and provides a reference for selecting an optimal portfolio according to the investor’s individual preferences. More risk-averse investors will prefer portfolios with lower volatility, while others may be willing to accept higher risk in exchange for higher expected returns.

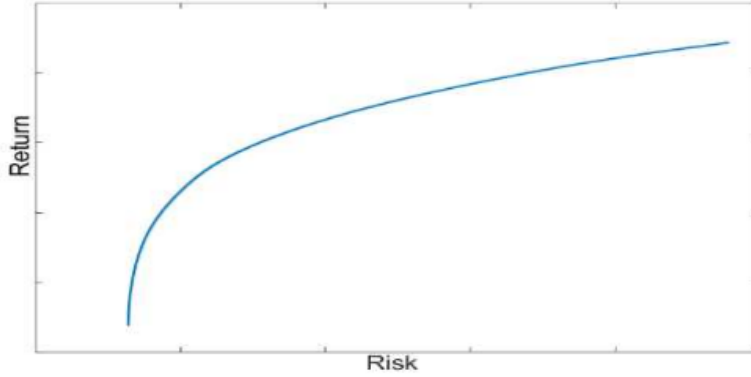


Figure 3: Efficient frontier plot example. Each point represents a feasible portfolio, while the curve shows the trade-offs between risk and return.

Portfolios that lie below the efficient frontier are inefficient, as there exist alternative portfolios that achieve higher returns for the same level of risk or lower risk for the same return.

5 S&P 500 Example

Using the historical daily returns of the selected assets, we estimated the expected return vector $\hat{\mu}$ by the sample mean and the covariance matrix $\hat{\Sigma}$ by the sample covariance matrix. These estimates were then used as inputs in the portfolio optimization models described in Section 4

We consider a small sample of 10 assets (Apple, Microsoft, NVIDIA, Amazon, Alphabet, Meta, JPMorgan, Exxon, Johnson & Johnson, Coca-Cola, Walmart), and bigger samples of 50, 100, and 300 assets for our analysis.

We used the Python library `yfinance`, which sources the data from Yahoo Finance, to obtain the daily adjusted closing prices for every company in the 500 S&P that had complete historical data of the prices of the assets from January 2019 to December 31 2024. A total of 480 companies were available and the first 300 from the Wikipedia's 500 S&P list of companies were used for the analysis.

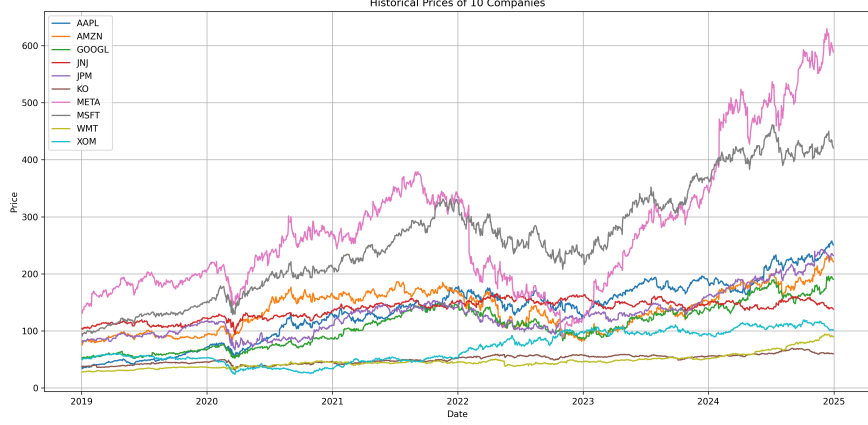


Figure 4: Selected companies' price over time graph (2019-2024)

We then calculate the returns using equation 2. To better visualize the performance over time, we compute the cumulative returns, which represent the growth of an initial investment of one dollar. The resulting plot is shown in Figure 5.

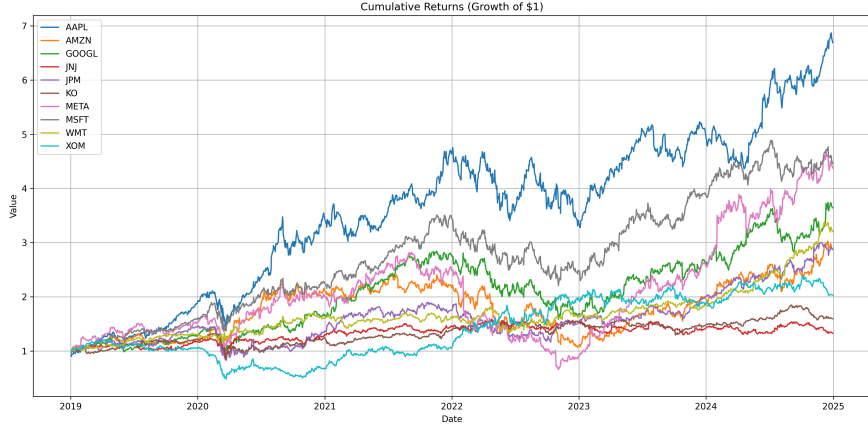


Figure 5: Cumulative returns of the selected S&P 500 companies, representing the growth of a one-dollar investment over the sample period.

5.1 Experimental Setup

We solved three models: the basic model, the budget-constrained model, and the diversification-constrained model. For the diversification model, we imposed upper bounds $w \leq u$ to prevent excessive concentration in a single asset. We then varied the risk target σ^{tar} over a grid of values to study the sensitivity of the optimal portfolios.

5.2 Results for a single target risk

For $\sigma^{tar} = 0.3084$, and for each model report:

Model	Expected Return	Risk	$\sum_i w_i$	# Active Assets
Basic	0.3653	0.3085	1	1 (AAPL)
Budget	0.3653	0.3085	1	1 (AAPL)
Diversification	0.3150	0.2862	1	4 (AAPL, GOOGL, META, MSFT)

Table 1: Comparison of optimal portfolios for a fixed target risk $\sigma_{tar} = 0.3084$.

For this small sample of assets, we can see that the Basic and budget model have the same weights, since the budget model invested all the capital. The diversification model spreaded the weight evenly in accordance to the 0.25 constraint of the weights.

We present the same table for 300 companies in the S&P:

Model	Expected Return	Risk	$\sum_i w_i$	# Active Assets
Basic	0.6785	0.5553	1	3
Budget	0.6785	0.5553	1	3
Diversification	0.6240	0.4589	1	4

Table 2: Comparison of optimal portfolios for a fixed target risk $\sigma_{tar} = 0.3084$.

Model	Basic	Budget	Diversification
ARES	NA	NA	0.25
AXON	0.37532	0.37532	0.25
BLDR	0.26405	0.26405	0.25
CVNA	0.36063	0.36063	0.25

Table 3: Comparison of weights across models.

5.3 Efficient frontier Sensitivity analysis

To analyze the impact of the risk constraint on the optimal portfolios, we varying the target volatility σ_{tar} over a range of values. For each value of σ_{tar} , we solve the optimization problem and record the corresponding optimal expected return and the portfolio risk of the solution.

This allows us to trace out the efficient frontier to represents the set of portfolios that achieve the highest expected return for a given level of risk.

We compute and compare the efficient frontiers for the basic model, the budget constrained model, and the diversification constrained model.

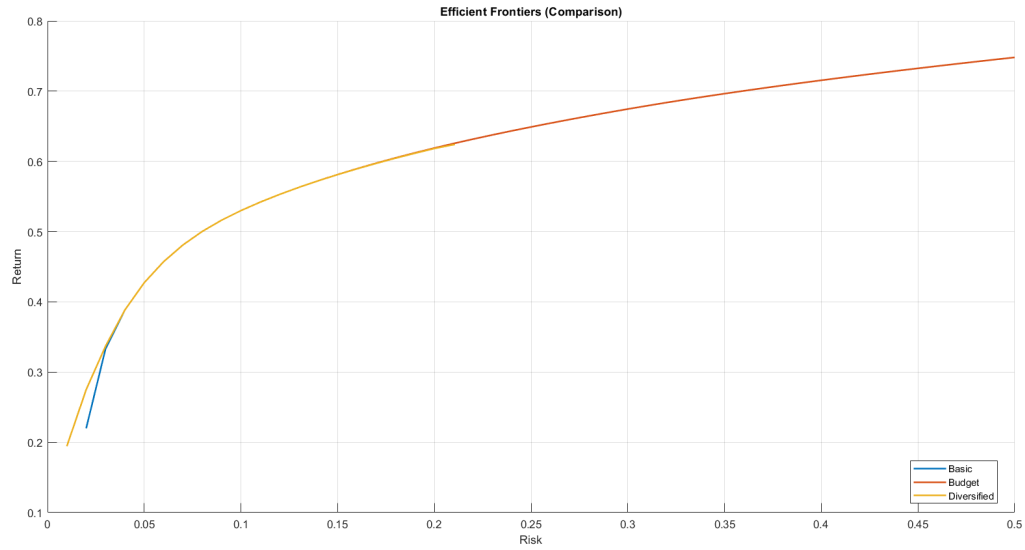


Figure 6: Here we have the frontier comparison between the 3 models, we can see that amongst them there is not a lot of difference for our data.

We can see, that as the target risk increases the 3 models seem to have almost the same results. Within the range of risk between .1 and .4, we do see the Basic model performing better.

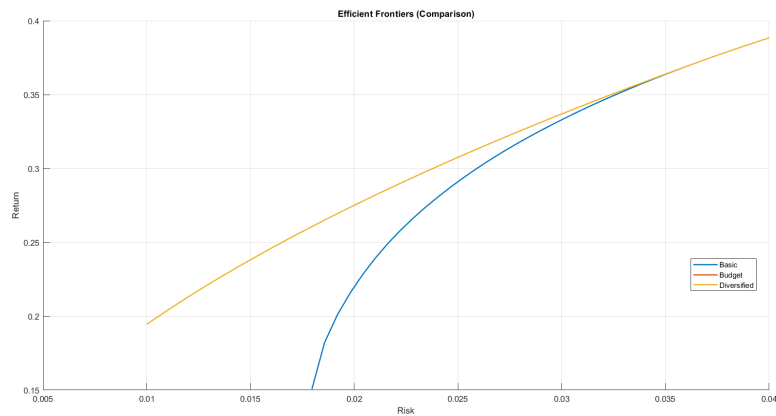


Figure 7: Frontier for a smaller range of risk

5.3.1 Sensitivity

To analyze how sensitive our model is we keep track of the dual variables as we vary the target risk value and then we plot the dual variable with respect to the realized risk.

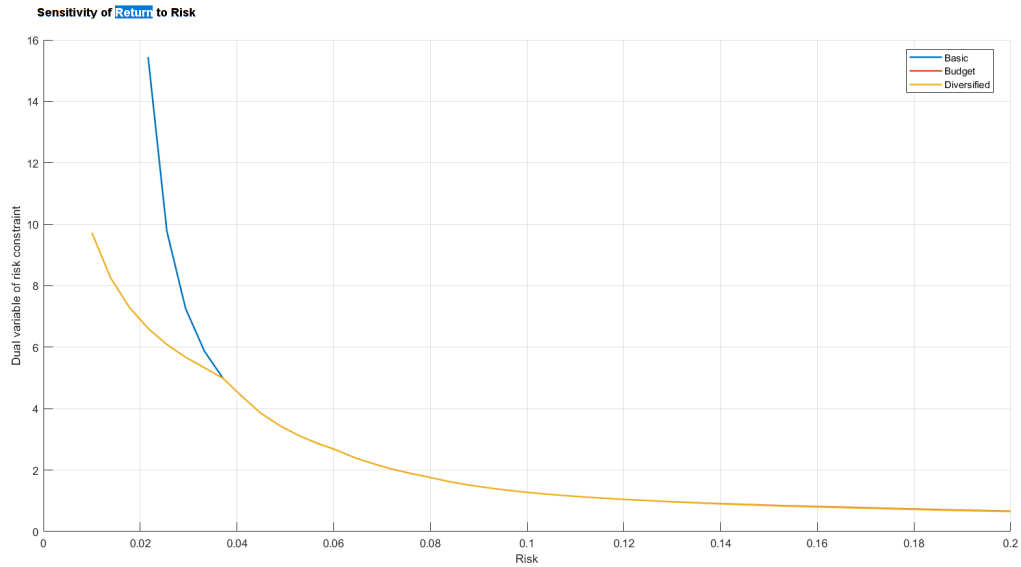


Figure 8: Plot of the dual variable given the target risk

We can see that for the basic model the dual variable starts with a big value, big in our problems context

6 Critique of the Model

The model assumes returns are fully characterized by mean and covariance. However:

- Returns are not normally distributed
- Extreme events are underestimated
- Temporal dependencies may exist

7 Conclusion

The mean-variance framework provides a tractable approach to portfolio optimization, though its assumptions limit its accuracy in real markets.